

## Problem

A load is modeled as a 100-mH inductor in parallel with a 12- $\Omega$  resistor. A capacitor is needed to be connected to the load so that the network is critically damped at 60 Hz. Calculate the size of the capacitor.

## Step-by-step solution

## Step 1 of 2

For the parallel connected RLC circuit to be critically damped, the value of the neper frequency must be equal to the resonant frequency. That is,

$$\alpha = \omega_0$$

Substitute  $\frac{1}{2RC}$  for  $\alpha$  and  $\frac{1}{\sqrt{LC}}$  for  $\omega_0$ .

$$\begin{aligned}\frac{1}{2RC} &= \frac{1}{\sqrt{LC}} \\ 2RC &= \sqrt{LC}\end{aligned}$$

Square the equation on both sides.

$$(2RC)^2 = (\sqrt{LC})^2$$

$$4R^2C^2 = LC$$

$$4R^2C = L$$

$$C = \frac{L}{4R^2} \dots\dots (1)$$

## Step 2 of 2

Substitute 12  $\Omega$  for  $R$  and 100 mH for  $L$  in equation (1).

$$\begin{aligned}C &= \frac{100 \text{ m}}{4(12)^2} \\ &= \frac{25 \text{ m}}{144} \\ &= 173.61 \mu\text{F}\end{aligned}$$

Therefore, the required value of the capacitance  $C$  is  $\boxed{173.61 \mu\text{F}}$ .